09/12/24 M/S

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[22ME51]

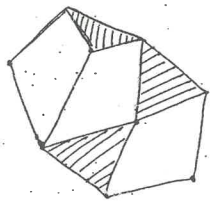
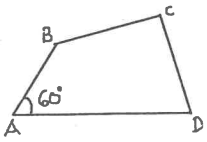
FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

THEORY OF MACHINES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define machine and mechanism	L1	CO1	**02**
	b.	Define completely and incompletely constrained motion	L1	CO1	**02**
	c.	List the different types of cam	L1	CO3	**02**
	d.	What is the function of governor?	L1	CO4	**02**
	e.	Define gyroscopic couple.	L1	CO4	**02**
		PART-B	**6X5=30**		
2.	**a.**	Determine the degree of freedom of the kinematic linkages shown in Fig. 2a.  Fig. 2a	L3	CO1	**05**
		OR			
	b.	Explain with a neat sketch hand pump mechanism.	L2	CO	**05**
3.	**a.**	In a pin jointed four bar mechanism shown in Fig 3a of AB= 300mm, BC=CD=360 mm and AD= 600 mm angle BAD = 60° the crank AB rotates at 100 rpm locate all the instantaneous centres and find the angular velocity of the link BC and velocity of C.  Fig. 3a	L3	CO2	**05**
		OR			
	b.	Differentiate between machine and mechanism.	L2	CO1	**05**
4.	**a.**	Explain the following types of cam, **i.** D-R-D cam **ii.** R-R-R cam	L2	CO3	**05**
		OR			
	b.	Explain different types of follower with neat sketch.	L2	CO3	**05**

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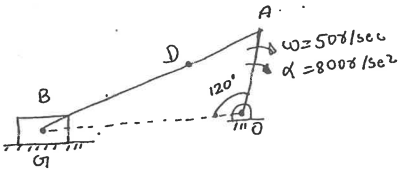
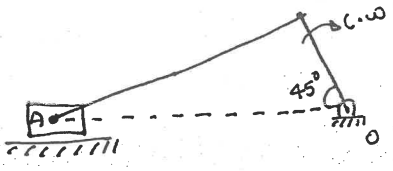
5.	a.	What is gyroscopic couple? Derive a relation for its magnitude.	L2	CO4	05
		OR			
	b.	Differentiate between working principle of governor and flywheel.	L3	CO4	05
6.	a.	Explain energy stored in the flywheel during motion.	L2	CO3	05
		OR			
	b.	What is Coriolis component? Derive the expression for the same.	L2	CO2	05
7.	a.	Explain maximum function energy using turning moment diagram.	L2	CO3	05
		OR			
	b.	List out the differences between flywheel and gyroscope and mention their applications.	L2	CO4	05
		PART-C	**6X10=60**		
8.	a.	Discuss Whitworth quick return motion mechanism with a neat sketch.	L2	CO1	10
		OR			
	b.	Fig 8b shows an engine mechanism. The dimensions of various links are as follows Crank OA=200mm connecting rod AB=600mm Distance of center of mass from crank end. AD=200 mm at the instant, the crank has an angular velocity of 50 rad/sec Clock wise and an angular acceleration of 800 rad/sec^2 Calculate the i) Velocity of point D ii) Angular acceleration of AB  Fig 8b	L4	CO2	10
9.	a.	Locate all the instantaneous centers of the slider crank mechanism as shown in Fig 9a. The lengths of the crank OB connecting rod AB are 100 mm and 400mm respectively if the crank rotates clockwise with an angular velocity of 10 rad/sec. Find angular velocity of AB and Velocity of slider A.  Fig. 9a	L4	CO2	10
		OR			
	b.	Sketch and explain oscillating cylinder engine mechanism and Oldhaum's coupling.	L2	CO1	10

10.	**a.**	Roller follower, in-line; lift = 25mm; base circle radius = 20mm; roller radius = 5mm; out stroke with UARM, for 120°cam rotation; dwell for 60° cam rotation; return stroke with UARM, for 90° cam rotation; dwell for the remaining period. Determine max. Velocity and acceleration during out stroke and return stroke if the cam rotates at 1200 rpm in clockwise direction.	L4	CO3	**10**
		OR			
	b.	Sketch and explain crank and slotted lever mechanism with its advantages and applications.	L2	CO1	**10**
11.	**a.**	The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8 m. It is found from the turning mount diagram that the fluctuation of energy is 56 kN-m. If the mean speed of the engine is 120 rpm find the maximum speeds.	L3	CO3	**10**
		OR			
	b.	Draw turning moment diagram of single cylinder engine, explain energy stored at each cycle.	L2	CO3	**10**
12.	**a.**	The motor of a marine having a mass of 1000 Kg and radius of gyration 300 mm rotates at 1550 rpm clockwise when looking from the bow. Determine the gyroscopic couple and its effect on the ship in the following cases. **i.** When the ship pitches with an angular velocity of 1 rad/sec when the a) Rising b) falling **ii.** When the ship is speeding at 40 km/hr and takes a right turn in a circular path of 200 m radius **iii.** When the ship rolls at certain instant, it has angular velocity of 0.5rad/sec when viewed from the stern	L4	CO4	**10**
		OR			
	b.	Draw the gyroscopic effect on aeroplane when aeroplane takes left turn rotor rotates clockwise direction viewed from rear end.	L3	CO4	**10**
13.	**a.**	Each arm of a porter governor is 200 mm long is pivoted on the axis of the governor. The radii of rotation of the balls at the minimum and maximum speeds are 120 mm and 160 mm respectively. The mass pf the sleeve is 24Kg and each ball is 4Kg. Find the range of the speed of the governor	L4	CO5	**10**
		OR			
	b.	Discuss the different ways of balancing of rotating masses.	L2	CO5	**10**